0707.4470 — formula report

213 inline formulas, 62 display equations, 0 tables, 12 unrecovered image regions.

1 row shows a REFINEMENT below, not the OCR reading: 1 verified against the source. Each is marked [refined] beside its identifier. The model is unchanged — the original reading is still what ‘latex‘ holds; this projection chose the refinement for it. Rows: 0707.4470_FO0175

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0707.4470_EQ0025	017	0.382 W|+1	`\begin{aligned} & F=E_{x} \mathrm{~d} x \wedge \mathrm{~d} t+E_{y} \mathrm{~d} y \wedge \mathrm{~d} t+E_{z} \mathrm{~d} z \wedge d t \\ & \quad+B_{x} \mathrm{~d} y \wedge \mathrm{~d} z+B_{y} \mathrm{~d} z \wedge \mathrm{~d} x+B_{z} \mathrm{~d} x \wedge \mathrm{~d} y \end{aligned}`	$F=E_{x} \mathrm{~d} x \wedge \mathrm{~d} t+E_{y} \mathrm{~d} y \wedge \mathrm{~d} t+E_{z} \mathrm{~d} z \wedge d t$ $\quad+B_{x} \mathrm{~d} y \wedge \mathrm{~d} z+B_{y} \mathrm{~d} z \wedge \mathrm{~d} x+B_{z} \mathrm{~d} x \wedge \mathrm{~d} y$	$F=E_x\,\mathrm{d}x\wedge\mathrm{d}t+E_y\,\mathrm{d}y\wedge\mathrm{d}t+E_z\,\mathrm{d}z\wedge d\,t$ $+B_x\,\mathrm{d}y\wedge\mathrm{d}z+B_y\,\mathrm{d}z\wedge\mathrm{d}x+B_z\,\mathrm{d}x\wedge\mathrm{d}y,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
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C|+40</td><td>\begin{aligned} & \frac{\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n+1}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}}{\Delta t}= \\ & \frac{\left.E_{y}\right|_{k, l+\frac{1}{2}, m+1} ^{n+\frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}}{\Delta z}-\frac{\left.E_{z}\right|_{k, l+1, m+\frac{1}{2}} ^{n+\frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}}{\Delta y} \\ & \frac{\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n+1}-\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}}{\Delta t}= \\ & \frac{\left.E_{z}\right|_{k+1, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}}{\Delta x}-\frac{\left.E_{x}\right|_{k+\frac{1}{2}, l, m+1} ^{n+\frac{1}{2}, m}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}}{\Delta z} \\ & \frac{\left.E_{x}\right|_{k+\frac{1}{2}, l+1, m} ^{n+\frac{1}{2}}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}}{\Delta y}-\frac{\left.E_{y}\right|_{k+1, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}}{\Delta x} \end{aligned}</td><td>$$\begin{aligned} & \frac{\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n+1}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}}{\Delta t}= \\ & \frac{\left.E_{y}\right|_{k, l+\frac{1}{2}, m+1} ^{n+\frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}}{\Delta z}-\frac{\left.E_{z}\right|_{k, l+1, m+\frac{1}{2}} ^{n+\frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}}{\Delta y} \\ & \frac{\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n+1}-\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}}{\Delta t}= \\ & \frac{\left.E_{z}\right|_{k+1, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}}{\Delta x}-\frac{\left.E_{x}\right|_{k+\frac{1}{2}, l, m+1} ^{n+\frac{1}{2}, m}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}}{\Delta z} \\ & \frac{\left.E_{x}\right|_{k+\frac{1}{2}, l+1, m} ^{n+\frac{1}{2}}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}}{\Delta y}-\frac{\left.E_{y}\right|_{k+1, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}}{\Delta x} \end{aligned}$$</td><td>$$\begin{aligned} & \frac{\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n+1}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}}{\Delta t}= \\ & \qquad \frac{\left.E_{y}\right|_{k, l+\frac{1}{2}, m+1} ^{n+\frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}}{\Delta z}-\frac{\left.E_{z}\right|_{k, l+1, m+\frac{1}{2}} ^{n+\frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}}{\Delta y} \\ & \frac{\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n+1}-\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}}{\Delta t}= \\ & \qquad \frac{\left.E_{z}\right|_{k+1, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}}{\Delta x}-\frac{\left.E_{x}\right|_{k+\frac{1}{2}, l, m+1} ^{n+\frac{1}{2}}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}}{\Delta z} \\ & \frac{\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n+1}-\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}}{\Delta t}= \\ & \qquad \frac{\left.E_{x}\right|_{k+\frac{1}{2}, l+1, m} ^{n+\frac{1}{2}}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}}{\Delta y}-\frac{\left.E_{y}\right|_{k+1, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}}{\Delta x} \end{aligned}$$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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● C|+22</td><td><code>\begin{aligned} & \frac{\left.D_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}-\left.D_{x}\right|_{k+\frac{1}{2}, l, m} ^{n-\frac{1}{2}}}{\Delta t}= \\ & \quad \frac{\left.H_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}-\left.H_{z}\right|_{k+\frac{1}{2}, l-\frac{1}{2}, m} ^{n}}{\Delta y}-\frac{\left.H_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}-\left.H_{y}\right|_{k+\frac{1}{2}, l, m-\frac{1}{2}} ^{n}}{\Delta z}-\left.J_{x}\right|_{k+\frac{1}{2}, l, m} ^{n} \\ & \quad \frac{\left.D_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}-\left.D_{y}\right|_{k, l+\frac{1}{2}, m} ^{n-\frac{1}{2}}}{\Delta t}= \\ & \quad \frac{\left.H_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}-\left.H_{x}\right|_{k, l+\frac{1}{2}, m-\frac{1}{2}} ^{n}}{\Delta z}-\frac{\left.H_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}-\left.H_{z}\right|_{k-\frac{1}{2}, l+\frac{1}{2}, m} ^{n}-\left.D_{z}\right|_{k, l, m+\frac{1}{2}} ^{n-\frac{1}{2}}}{\Delta x}=\left.J_{y}\right|_{k, l+\frac{1}{2}, m} ^{n} & \quad \frac{\left.H_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}-\left.H_{y}\right|_{k-\frac{1}{2}, l, m+\frac{1}{2}} ^{n}}{\Delta x}-\frac{\left.H_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}-\left.H_{x}\right|_{k, l-\frac{1}{2}, m+\frac{1}{2}} ^{n}}{\Delta y}-\left.J_{z}\right|_{k, l, m+\frac{1}{2}} ^{n} \end{aligned}</code></td><td>$\begin{aligned} & \frac{\left.D_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}-\left.D_{x}\right|_{k+\frac{1}{2}, l, m} ^{n-\frac{1}{2}}}{\Delta t}= \\ & \quad \frac{\left.H_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}-\left.H_{z}\right|_{k+\frac{1}{2}, l-\frac{1}{2}, m} ^{n}}{\Delta y}-\frac{\left.H_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}-\left.H_{y}\right|_{k+\frac{1}{2}, l, m-\frac{1}{2}} ^{n}}{\Delta z}-\left.J_{x}\right|_{k+\frac{1}{2}, l, m} ^{n} \\ & \quad \frac{\left.D_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}-\left.D_{y}\right|_{k, l+\frac{1}{2}, m} ^{n-\frac{1}{2}}}{\Delta t}= \\ & \quad \frac{\left.H_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}-\left.H_{x}\right|_{k, l+\frac{1}{2}, m-\frac{1}{2}} ^{n}}{\Delta z}-\frac{\left.H_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}-\left.H_{z}\right|_{k-\frac{1}{2}, l+\frac{1}{2}, m} ^{n}-\left.D_{z}\right|_{k, l, m+\frac{1}{2}} ^{n-\frac{1}{2}}}{\Delta x}=\left.J_{y}\right|_{k, l+\frac{1}{2}, m} ^{n} & \quad \frac{\left.H_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}-\left.H_{y}\right|_{k-\frac{1}{2}, l, m+\frac{1}{2}} ^{n}}{\Delta x}-\frac{\left.H_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}-\left.H_{x}\right|_{k, l-\frac{1}{2}, m+\frac{1}{2}} ^{n}}{\Delta y}-\left.J_{z}\right|_{k, l, m+\frac{1}{2}} ^{n} \end{aligned}$</td><td>$\begin{aligned} & \frac{D_x\big|^{n+\frac{1}{2}}_{k+\frac{1}{2},l,m} - D_x\big|^{n-\frac{1}{2}}_{k+\frac{1}{2},l,m}}{\Delta t} = \\ & \quad \frac{H_z\big|^{n}_{k+\frac{1}{2},l+\frac{1}{2},m} - H_z\big|^{n}_{k+\frac{1}{2},l-\frac{1}{2},m}}{\Delta y} - \frac{H_y\big|^{n}_{k+\frac{1}{2},l,m+\frac{1}{2}} - H_y\big|^{n}_{k+\frac{1}{2},l,m-\frac{1}{2}}}{\Delta z} - J_x\big|^{n}_{k+\frac{1}{2},l,m} \\ & \frac{D_y\big|^{n+\frac{1}{2}}_{k,l+\frac{1}{2},m} - D_y\big|^{n-\frac{1}{2}}_{k,l+\frac{1}{2},m}}{\Delta t} = \\ & \quad \frac{H_x\big|^{n}_{k,l+\frac{1}{2},m+\frac{1}{2}} - H_x\big|^{n}_{k,l+\frac{1}{2},m-\frac{1}{2}}}{\Delta z} - \frac{H_z\big|^{n}_{k+\frac{1}{2},l+\frac{1}{2},m} - H_z\big|^{n}_{k-\frac{1}{2},l+\frac{1}{2},m}}{\Delta x} - J_y\big|^{n}_{k,l+\frac{1}{2},m} \\ & \frac{D_z\big|^{n+\frac{1}{2}}_{k,l,m+\frac{1}{2}} - D_z\big|^{n-\frac{1}{2}}_{k,l,m+\frac{1}{2}}}{\Delta t} = \\ & \quad \frac{H_y\big|^{n}_{k+\frac{1}{2},l,m+\frac{1}{2}} - H_y\big|^{n}_{k-\frac{1}{2},l,m+\frac{1}{2}}}{\Delta x} - \frac{H_x\big|^{n}_{k,l+\frac{1}{2},m+\frac{1}{2}} - H_x\big|^{n}_{k,l-\frac{1}{2},m+\frac{1}{2}}}{\Delta y} - J_z\big|^{n}_{k,l,m+\frac{1}{2}} \end{aligned}$</td></tr>
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● K|+0</td><td><code>\int_{\sigma} \mathrm{d} \alpha=\int_{\partial \sigma} \alpha .</code></td><td>$\int_{\sigma} \mathrm{d} \alpha=\int_{\partial \sigma} \alpha .$</td><td>$\int_{\sigma} \mathrm{d}\alpha = \int_{\partial\sigma} \alpha .$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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● W|+0</td><td><code>\begin{aligned} x z t \text {-face : } & -\left(\left.E_{x}\right|_{k+\frac{1}{2}, l, m+1} ^{n+ \frac{1}{2}}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}\right) \Delta x \Delta t \\ & + \left(\left.E_{z}\right|_{k+1, l, m+\frac{1}{2}} ^{n+ \frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}\right) \Delta z \Delta t \\ & -\left(\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n+1}-\left.B_{y}\right|_{k+\frac{1}{2}, l, m+ \frac{1}{2}} ^{n}\right) \Delta x \Delta z \end{aligned}</code></td><td>$$\begin{aligned} x z t \text {-face : } & -\left(\left.E_{x}\right|_{k+\frac{1}{2}, l, m+1} ^{n+ \frac{1}{2}}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}\right) \Delta x \Delta t \\ & + \left(\left.E_{z}\right|_{k+1, l, m+\frac{1}{2}} ^{n+ \frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}\right) \Delta z \Delta t \\ & -\left(\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n+1}-\left.B_{y}\right|_{k+\frac{1}{2}, l, m+ \frac{1}{2}} ^{n}\right) \Delta x \Delta z \end{aligned}$$</td><td>$$\begin{aligned} x z t \text {-face : } & -\left(\left.E_{x}\right|_{k+\frac{1}{2}, l, m+1} ^{n+ \frac{1}{2}}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}\right) \Delta x \Delta t \\ & + \left(\left.E_{z}\right|_{k+1, l, m+\frac{1}{2}} ^{n+ \frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}\right) \Delta z \Delta t \\ & -\left(\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n+1}-\left.B_{y}\right|_{k+\frac{1}{2}, l, m+ \frac{1}{2}} ^{n}\right) \Delta x \Delta z \end{aligned}$$</td></tr>
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● W|+1</td><td><code>\begin{aligned} x y z \text {-face : } & \left(\left.B_{x}\right|_{k+1, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}\right) \Delta y \Delta z \\ & +\left(\left.B_{y} \right|_{k+\frac{1}{2}, l+1, m+\frac{1}{2}} ^{n}- \left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}\right) \Delta x \Delta z \\ & +\left(\left.B_{z} \right|_{k+\frac{1}{2}, l+\frac{1}{2}, m+1} ^{n}- \left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}\right) \Delta x \Delta y \end{aligned}</code></td><td>$$\begin{aligned} x y z \text {-face : } & \left(\left.B_{x}\right|_{k+1, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}\right) \Delta y \Delta z \\ & +\left(\left.B_{y} \right|_{k+\frac{1}{2}, l+1, m+\frac{1}{2}} ^{n}- \left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}\right) \Delta x \Delta z \\ & +\left(\left.B_{z} \right|_{k+\frac{1}{2}, l+\frac{1}{2}, m+1} ^{n}- \left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}\right) \Delta x \Delta y \end{aligned}$$</td><td>$$\begin{aligned} x y z \text {-face : } & \left(\left.B_{x}\right|_{k+1, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}\right) \Delta y \Delta z \\ & +\left(\left.B_{y} \right|_{k+\frac{1}{2}, l+1, m+\frac{1}{2}} ^{n}- \left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}\right) \Delta x \Delta z \\ & +\left(\left.B_{z} \right|_{k+\frac{1}{2}, l+\frac{1}{2}, m+1} ^{n}- \left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}\right) \Delta x \Delta y \end{aligned}$$</td></tr>
<tr><td>0707.4470_EQ0026</td><td>018</td><td>■ 0.856
● N|+1</td><td><code>\begin{gathered} x t \text {-face : }\left.E_{x} \right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}} \Delta x \Delta t \\ y t \text {-face : }\left.E_{y}\right|_{k, l+ \frac{1}{2}, m} ^{n+\frac{1}{2}} \Delta y \Delta t \\ z t \text {-face : }\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}} \Delta z \Delta t \\ y z \text {-face : }\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n} \Delta y \Delta z \\ x z \text {-face : }\left.B_{y} \right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n} \Delta z \Delta x \\ x y \text {-face : }\left.B_{z}\right|_{k+ \frac{1}{2}, l+\frac{1}{2}, m} ^{n} \Delta x \Delta y \end{gathered}</code></td><td>$$\begin{gathered} x t \text {-face : }\left.E_{x} \right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}} \Delta x \Delta t \\ y t \text {-face : }\left.E_{y}\right|_{k, l+ \frac{1}{2}, m} ^{n+\frac{1}{2}} \Delta y \Delta t \\ z t \text {-face : }\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}} \Delta z \Delta t \\ y z \text {-face : }\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n} \Delta y \Delta z \\ x z \text {-face : }\left.B_{y} \right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n} \Delta z \Delta x \\ x y \text {-face : }\left.B_{z}\right|_{k+ \frac{1}{2}, l+\frac{1}{2}, m} ^{n} \Delta x \Delta y \end{gathered}$$</td><td>$$\begin{gathered} x t \text {-face : }\left.E_{x} \right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}} \Delta x \Delta t \\ y t \text {-face : }\left.E_{y}\right|_{k, l+ \frac{1}{2}, m} ^{n+\frac{1}{2}} \Delta y \Delta t \\ z t \text {-face : }\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}} \Delta z \Delta t \\ y z \text {-face : }\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n} \Delta y \Delta z \\ x z \text {-face : }\left.B_{y} \right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n} \Delta z \Delta x \\ x y \text {-face : }\left.B_{z}\right|_{k+ \frac{1}{2}, l+\frac{1}{2}, m} ^{n} \Delta x \Delta y. \end{gathered}$$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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<tr><td>0707.4470_EQ0030</td><td>018</td><td>■ 0.881
● W|+0</td><td><code>\begin{aligned} y z t \text {-face: } & -\left(\left.E_{y}\right|_{k, l+\frac{1}{2}, m+1} ^{n+ \frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}\right) \Delta y \Delta t \\ & + \left(\left.E_{z}\right|_{k, l+1, m+\frac{1}{2}} ^{n+ \frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}\right) \Delta z \Delta t \\ & +\left(\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n+1}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+ \frac{1}{2}} ^{n}\right) \Delta y \Delta z \end{aligned}</code></td><td>$$\begin{aligned} yzt\text{-face: } & -\left(\left.E_{y}\right|_{k, l+\frac{1}{2}, m+1} ^{n+\frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}\right) \Delta y \Delta t \\ & + \left(\left.E_{z}\right|_{k, l+1, m+\frac{1}{2}} ^{n+\frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}\right) \Delta z \Delta t \\ & +\left(\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n+1}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}\right) \Delta y \Delta z \end{aligned}$$</td><td>$$\begin{aligned} yzt\text{-face}: & -\left(\left.E_{y}\right|_{k, l+\frac{1}{2}, m+1} ^{n+\frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}\right) \Delta y \Delta t \\ & + \left(\left.E_{z}\right|_{k, l+1, m+\frac{1}{2}} ^{n+\frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}\right) \Delta z \Delta t \\ & +\left(\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n+1}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}\right) \Delta y \Delta z \end{aligned}$$</td></tr>
<tr><td>0707.4470_ EQ0058 (5.3)</td><td>032</td><td>■ 0.887
● N|+2</td><td><code>\begin{aligned} \mathbf{d} S[A] \cdot \alpha & =\int_{X} \left(\mathrm{~d}_{t} \alpha \wedge D-\mathrm{d}_{s} \alpha \wedge H \wedge \mathrm{~d} t+\alpha \wedge J \wedge \mathrm{~d} t\right) \\ & =\int_{X} \alpha \wedge\left(\mathrm{~d}_{t} D-\mathrm{d}_{s} H \wedge \mathrm{~d} t+J \wedge \mathrm{~d} t\right) \end{aligned}</code></td><td>$$\begin{aligned} \mathbf{d} S[A] \cdot \alpha & =\int_{X} \left(\mathrm{~d}_{t} \alpha \wedge D-\mathrm{d}_{s} \alpha \wedge H \wedge \mathrm{~d} t+\alpha \wedge J \wedge \mathrm{~d} t\right) \\ & =\int_{X} \alpha \wedge\left(\mathrm{~d}_{t} D-\mathrm{d}_{s} H \wedge \mathrm{~d} t+J \wedge \mathrm{~d} t\right) \end{aligned}$$</td><td>$$\begin{aligned} \mathbf{d} S[A] \cdot \alpha & =\int_{X} (\mathrm{d}_{t} \alpha \wedge D-\mathrm{d}_{s} \alpha \wedge H \wedge \mathrm{d} t+\alpha \wedge J \wedge \mathrm{d} t) \\ & =\int_{X} \alpha \wedge(\mathrm{d}_{t} D-\mathrm{d}_{s} H \wedge \mathrm{d} t+J \wedge \mathrm{d} t). \end{aligned}$$</td></tr>
<tr><td>0707.4470_EQ0059</td><td>032</td><td>■ 0.892
● K|+0</td><td><code>\mathbf{d} S[A] \cdot \alpha=\left.\int_{\Sigma} \alpha \wedge D\right|_{t_{0}} ^{t_{f}},</code></td><td>$$\mathbf{d} S[A] \cdot \alpha=\left.\int_{\Sigma} \alpha \wedge D\right|_{t_{0}} ^{t_{f}},$$</td><td>$$\mathbf{d} S[A] \cdot \alpha=\left.\int_{\Sigma} \alpha \wedge D\right|_{t_{0}} ^{t_{f}},$$</td></tr>
<tr><td>0707.4470_ EQ0016 (2.5)</td><td>008</td><td>■ 0.911
● N|-1</td><td><code>\begin{aligned} \mathbf{d} S[A, F, G] \cdot(\alpha, \phi, \gamma) & =\int_{X}[-\phi \wedge * F+\alpha \wedge \mathscr{J}+(\phi-\mathrm{d} \alpha) \wedge G+(F- \mathrm{d} A) \wedge \gamma] \\ & =\int_{X}[\alpha \wedge(\mathscr{J}-\mathrm{d} G)+\phi \wedge(G-* F)+(F- \mathrm{d} A) \wedge \gamma] \end{aligned}</code></td><td>$$\begin{aligned} \mathbf{d} S[A, F, G] \cdot(\alpha, \phi, \gamma) & =\int_{X}[-\phi \wedge * F+\alpha \wedge \mathscr{J}+(\phi-\mathrm{d} \alpha) \wedge G+(F-\mathrm{d} A) \wedge \gamma] \\ & =\int_{X}[\alpha \wedge(\mathscr{J}-\mathrm{d} G)+\phi \wedge(G-* F)+(F-\mathrm{d} A) \wedge \gamma] \end{aligned}$$</td><td>$$\begin{aligned} \mathbf{d} S[A, F, G] \cdot(\alpha, \phi, \gamma) & =\int_{X}\left[-\phi \wedge * F+\alpha \wedge \mathscr{J}+(\phi-\mathrm{d} \alpha) \wedge G+(F-\mathrm{d} A) \wedge \gamma\right] \\ & =\int_{X}\left[\alpha \wedge(\mathscr{J}-\mathrm{d} G)+\phi \wedge(G-* F)+(F-\mathrm{d} A) \wedge \gamma\right]. \end{aligned}$$</td></tr>
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Conf. MathPix confidence: ■ $\geq$0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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W|+0</td><td>\begin{aligned} x y t \text {-face : } & -\left(\left.E_{x}\right|_{k+\frac{1}{2}, l+1, m} ^{n+ \frac{1}{2}}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}\right) \Delta x \Delta t \\ & + \left(\left.E_{y}\right|_{k+1, l+\frac{1}{2}, m} ^{n+ \frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}\right) \Delta y \Delta t \\ & +\left(\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n+1}-\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}\right) \Delta x \Delta y \end{aligned}</td><td>$$\begin{aligned} xyt\text{-face}: & -\left(\left.E_{x}\right|_{k+\frac{1}{2}, l+1, m}^{n+\frac{1}{2}}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}}\right) \Delta x \Delta t \\ & +\left(\left.E_{y}\right|_{k+1, l+\frac{1}{2}, m}^{n+\frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}}\right) \Delta y \Delta t \\ & +\left(\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m}^{n+1}-\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m}^{n}\right) \Delta x \Delta y \end{aligned}$$</td><td>$$\begin{aligned} xyt\text{-face}: & -\left(\left.E_{x}\right|_{k+\frac{1}{2}, l+1, m}^{n+\frac{1}{2}}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}}\right) \Delta x \Delta t \\ & +\left(\left.E_{y}\right|_{k+1, l+\frac{1}{2}, m}^{n+\frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}}\right) \Delta y \Delta t \\ & +\left(\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m}^{n+1}-\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m}^{n}\right) \Delta x \Delta y \end{aligned}$$</td></tr>
<tr><td>0707.4470_ EQ0036 (4.2)</td><td>020</td><td>0.951
W|+1</td><td>\begin{aligned} & \frac{\left.D_{x}\right|_{k+ \frac{1}{2}, l, m} ^{n+\frac{1}{2}}-\left.D_{x} \right|_{k-\frac{1}{2}, l, m} ^{n+\frac{1}{2}}}{ \Delta x}+\frac{\left.D_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}-\left.D_{y}\right|_{k, l- \frac{1}{2}, m} ^{n+\frac{1}{2}}}{\Delta y} \\ & \quad+ \frac{\left.D_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+ \frac{1}{2}}-\left.D_{z}\right|_{k, l, m-\frac{1}{2}} ^{n+\frac{1}{2}}}{\Delta z}=\left.\rho\right|_{k, l, m} ^{n+\frac{1}{2}} . \end{aligned}</td><td>$$\begin{aligned} & \frac{\left.D_{x}\right|_{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}}-\left.D_{x}\right|_{k-\frac{1}{2}, l, m}^{n+\frac{1}{2}}}{\Delta x}+\frac{\left.D_{y}\right|_{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}}-\left.D_{y}\right|_{k, l-\frac{1}{2}, m}^{n+\frac{1}{2}}}{\Delta y} \\ & \quad+\frac{\left.D_{z}\right|_{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}}-\left.D_{z}\right|_{k, l, m-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z}=\left.\rho\right|_{k, l, m}^{n+\frac{1}{2}} . \end{aligned}$$</td><td>$$\begin{aligned} & \frac{\left.D_{x}\right|_{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}}-\left.D_{x}\right|_{k-\frac{1}{2}, l, m}^{n+\frac{1}{2}}}{\Delta x}+\frac{\left.D_{y}\right|_{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}}-\left.D_{y}\right|_{k, l-\frac{1}{2}, m}^{n+\frac{1}{2}}}{\Delta y} \\ & \quad+\frac{\left.D_{z}\right|_{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}}-\left.D_{z}\right|_{k, l, m-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z}=\left.\rho\right|_{k, l, m}^{n+\frac{1}{2}} . \end{aligned}$$</td></tr>
<tr><td>0707.4470_EQ0023</td><td>013</td><td>0.955
N|+0</td><td>\begin{aligned} (\alpha, \beta) & =\sum_{\sigma^{k}} \kappa(\sigma)\binom{n}{k} \frac{|\mathrm{CH}(\sigma, * \sigma)|}{|\sigma|^{2}}\langle\alpha, \sigma\rangle \langle\beta, \sigma\rangle \\ & =\sum_{\sigma^{k}} \kappa(\sigma) \frac{|* \sigma|}{|\sigma|}\langle\alpha, \sigma\rangle\langle\beta, \sigma\rangle \end{aligned}</td><td>$$\begin{aligned} (\alpha, \beta) & =\sum_{\sigma^{k}} \kappa(\sigma)\binom{n}{k} \frac{|\mathrm{CH}(\sigma, * \sigma)|}{|\sigma|^{2}}\langle\alpha, \sigma\rangle\langle\beta, \sigma\rangle \\ & =\sum_{\sigma^{k}} \kappa(\sigma) \frac{|* \sigma|}{|\sigma|}\langle\alpha, \sigma\rangle\langle\beta, \sigma\rangle \end{aligned}$$</td><td>$$\begin{aligned} (\alpha, \beta) & =\sum_{\sigma^{k}} \kappa(\sigma)\binom{n}{k} \frac{|\mathrm{CH}(\sigma, * \sigma)|}{|\sigma|^{2}}\langle\alpha, \sigma\rangle\langle\beta, \sigma\rangle \\ & =\sum_{\sigma^{k}} \kappa(\sigma) \frac{|* \sigma|}{|\sigma|}\langle\alpha, \sigma\rangle\langle\beta, \sigma\rangle \end{aligned}$$</td></tr>
<tr><td>0707.4470_EQ0008</td><td>007</td><td>0.963
N|-2</td><td>\mathscr{L}=-\frac{1}{2} \mathrm{~d} A \wedge * \mathrm{~d} A+A \wedge \mathscr{J},</td><td>$$\mathscr{L}=-\frac{1}{2} \mathrm{~d} A \wedge * \mathrm{~d} A+A \wedge \mathscr{J},$$</td><td>$$\mathscr{L}=-\frac{1}{2} \mathrm{d} A \wedge * \mathrm{d} A+A \wedge \mathscr{J},$$</td></tr>
<tr><td>0707.4470_ EQ0012 (2.3)</td><td>008</td><td>0.982
N|+0</td><td>\begin{aligned} & \mathrm{d} F=0 \\ & \mathrm{~d} G= \mathscr{J} \end{aligned}</td><td>$$\begin{aligned} & \mathrm{d} F=0 \\ & \mathrm{~d} G=\mathscr{J} \end{aligned}$$</td><td>$$\begin{aligned} & \mathrm{d} F=0 \\ & \mathrm{d} G=\mathscr{J} \end{aligned}$$</td></tr>
<tr><td>0707.4470_EQ0060</td><td>032</td><td>0.982
K|+0</td><td>\mathbf{d} S[A] \cdot \mathrm{d}_{s} f=\left.\int_{ \Sigma} \mathrm{d}_{s} f \wedge D\right|_{t_{0}} ^{t_{f}}=-\left.\int_{\Sigma} f \wedge \mathrm{~d}_{s} D \right|_{t_{0}} ^{t_{f}}</td><td>$$\mathbf{d} S[A] \cdot \mathrm{d}_{s} f=\left.\int_{\Sigma} \mathrm{d}_{s} f \wedge D\right|_{t_{0}}^{t_{f}}=-\left.\int_{\Sigma} f \wedge \mathrm{~d}_{s} D\right|_{t_{0}}^{t_{f}}$$</td><td>$$\mathbf{d} S[A] \cdot \mathrm{d}_{s} f=\left.\int_{\Sigma} \mathrm{d}_{s} f \wedge D\right|_{t_{0}}^{t_{f}}=-\left.\int_{\Sigma} f \wedge \mathrm{d}_{s} D\right|_{t_{0}}^{t_{f}}$$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0707.4470_EQ0046 (4.4)	024	0.988 · W|+0	`\frac{D_{e}^{m+1 / 2}-D_{e}^{m-1 / 2}}{t_{e}^{m+1 / 2}-t_{e}^{m-1 / 2}}=\mathrm{d}_{1}^{T}\left(H_{\sigma}^{n} \nVdash\left\{t_{\sigma}^{n}=t_{e}^{m}\right\}\right)-J_{e}^{m},`	$\frac{D_{e}^{m+1 / 2}-D_{e}^{m-1 / 2}}{t_{e}^{m+1 / 2}-t_{e}^{m-1 / 2}}=\mathrm{d}_{1}^{T}\left(H_{\sigma}^{n} \nVdash\left\{t_{\sigma}^{n}=t_{e}^{m}\right\}\right)-J_{e}^{m},$	$\frac{D_{e}^{m+1/2}-D_{e}^{m-1/2}}{t_{e}^{m+1/2}-t_{e}^{m-1/2}}=\mathrm{d}_{1}^{T}\left(H_{\sigma}^{n} \nVdash_{\{t_{\sigma}^{n}=t_{e}^{m}\}}\right)-J_{e}^{m},$
0707.4470_EQ0002	006	0.996 · N|-1	`\oint_{C} \mathbf{E} \cdot \mathrm{~d} \mathbf{l}=-\frac{\mathrm{d}}{\mathrm{~d} t} \int_{S} \mathbf{B} \cdot \mathrm{~d} \mathbf{A},`	$\oint_{C} \mathbf{E} \cdot \mathrm{~d} \mathbf{l}=-\frac{\mathrm{d}}{\mathrm{~d} t} \int_{S} \mathbf{B} \cdot \mathrm{~d} \mathbf{A},$	$\oint_{C} \mathbf{E} \cdot \mathrm{d} \mathbf{l}=-\frac{\mathrm{d}}{\mathrm{d} t} \int_{S} \mathbf{B} \cdot \mathrm{d} \mathbf{A},$
0707.4470_EQ0013 (2.4)	008	0.996 · N|+0	`\begin{aligned} \nabla \times \mathbf{E}+\partial_{t} \mathbf{B} & =0 \\ \nabla \cdot \mathbf{B} & =0 . \end{aligned}`	$\begin{aligned} \nabla \times \mathbf{E}+\partial_{t} \mathbf{B} & =0 \\ \nabla \cdot \mathbf{B} & =0 . \end{aligned}$	$\begin{aligned} \nabla \times \mathbf{E}+\partial_{t} \mathbf{B} & =0 \\ \nabla \cdot \mathbf{B} & =0 . \end{aligned}$
0707.4470_EQ0001	005	0.997 · W|+1	`\begin{aligned} & E=E_{x} \mathrm{~d} x+E_{y} \mathrm{~d} y+E_{z} \mathrm{~d} z \\ & B=B_{x} \mathrm{~d} y \wedge \mathrm{~d} z+B_{y} \mathrm{~d} z \wedge \mathrm{~d} x+B_{z} \mathrm{~d} x \wedge \mathrm{~d} y \end{aligned}`	$\begin{aligned} & E=E_{x} \mathrm{~d} x+E_{y} \mathrm{~d} y+E_{z} \mathrm{~d} z \\ & B=B_{x} \mathrm{~d} y \wedge \mathrm{~d} z+B_{y} \mathrm{~d} z \wedge \mathrm{~d} x+B_{z} \mathrm{~d} x \wedge \mathrm{~d} y \end{aligned}$	$\begin{aligned} & E=E_{x} \mathrm{d} x+E_{y} \mathrm{d} y+E_{z} \mathrm{d} z \\ & B=B_{x} \mathrm{d} y \wedge \mathrm{d} z+B_{y} \mathrm{d} z \wedge \mathrm{d} x+B_{z} \mathrm{d} x \wedge \mathrm{d} y, \end{aligned}$
0707.4470_EQ0047	029	0.997 · N|-2	`\mathscr{L}_{d}=-\frac{1}{2} \mathrm{~d} A \wedge * \mathrm{~d} A+A \wedge \mathscr{J},`	$\mathscr{L}_{d}=-\frac{1}{2} \mathrm{~d} A \wedge * \mathrm{~d} A+A \wedge \mathscr{J},$	$\mathscr{L}_{d}=-\frac{1}{2} \mathrm{d} A \wedge * \mathrm{d} A+A \wedge \mathscr{J},$
0707.4470_EQ0015 (2.7)	008	0.999 · N|-1	`S[A, F, G]=\int_{X}\left[-\frac{1}{2} F \wedge * F+A \wedge \mathscr{J}+(F-\mathrm{d} A) \wedge G\right] .`	$S[A, F, G]=\int_{X}\left[-\frac{1}{2} F \wedge * F+A \wedge \mathscr{J}+(F-\mathrm{d} A) \wedge G\right] .$	$S[A, F, G]=\int_{X}\left[-\frac{1}{2} F \wedge * F+A \wedge \mathscr{J}+(F-\mathrm{d} A) \wedge G\right] .$
0707.4470_EQ0010	007	1.000 · N|-2	`\begin{aligned} \mathbf{d} S[A] \cdot \alpha & =\left. \frac{\mathrm{d}}{\mathrm{~d} \epsilon}\right|_{\epsilon=0} S[A+\epsilon \alpha] \\ & =\int_{X}(-\mathrm{d} \alpha \wedge * \mathrm{~d} A+\alpha \wedge \mathscr{J}) \\ & =\int_{X} \alpha \wedge(-\mathrm{d} * \mathrm{~d} A+\mathscr{J}), \end{aligned}`	$\begin{aligned} \mathbf{d} S[A] \cdot \alpha & =\left. \frac{\mathrm{d}}{\mathrm{~d} \epsilon}\right\vert_{\epsilon=0} S[A+\epsilon \alpha] \\ & =\int_{X}(-\mathrm{d} \alpha \wedge * \mathrm{~d} A+\alpha \wedge \mathscr{J}) \\ & =\int_{X} \alpha \wedge(-\mathrm{d} * \mathrm{~d} A+\mathscr{J}), \end{aligned}$	$\begin{aligned} \mathbf{d} S[A] \cdot \alpha & =\left. \frac{\mathrm{d}}{\mathrm{d} \epsilon}\right\vert_{\epsilon=0} S[A+\epsilon \alpha] \\ & =\int_{X}(-\mathrm{d} \alpha \wedge * \mathrm{d} A+\alpha \wedge \mathscr{J}) \\ & =\int_{X} \alpha \wedge(-\mathrm{d} * \mathrm{d} A+\mathscr{J}), \end{aligned}$
0707.4470_EQ0033 (4.1)	019	1.000 · W|+2	`\begin{aligned} & \frac{\left.B_{x}\right|_{k+1, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}}{\Delta x}+\frac{\left.B_{y}\right|_{k+\frac{1}{2}, l+1, m+\frac{1}{2}} ^{n}-\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}}{\Delta y} \\ & \quad+\frac{\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m+1} ^{n}-\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}}{\Delta z}=0 . \end{aligned}`	$\begin{aligned} & \frac{\left.B_{x}\right\vert_{k+1, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}-\left.B_{x}\right\vert_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}}{\Delta x}+\frac{\left.B_{y}\right\vert_{k+\frac{1}{2}, l+1, m+\frac{1}{2}} ^{n}-\left.B_{y}\right\vert_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}}{\Delta y} \\ & \quad+\frac{\left.B_{z}\right\vert_{k+\frac{1}{2}, l+\frac{1}{2}, m+1} ^{n}-\left.B_{z}\right\vert_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}}{\Delta z}=0 . \end{aligned}$	$\begin{aligned} & \frac{\left.B_{x}\right\vert_{k+1, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}-\left.B_{x}\right\vert_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}}{\Delta x}+\frac{\left.B_{y}\right\vert_{k+\frac{1}{2}, l+1, m+\frac{1}{2}} ^{n}-\left.B_{y}\right\vert_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}}{\Delta y} \\ & \quad+\frac{\left.B_{z}\right\vert_{k+\frac{1}{2}, l+\frac{1}{2}, m+1} ^{n}-\left.B_{z}\right\vert_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}}{\Delta z}=0 . \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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● W|-2</td><td><code>\mathbf{d} S_{d}[A] \cdot \alpha=\langle-\mathrm{d} \alpha \wedge * \mathrm{~d} A+\alpha \wedge \mathscr{J}, K\rangle=\langle\alpha \wedge(-\mathrm{d} * \mathrm{~d} A+\mathscr{J}), K\rangle .</code></td><td>$\mathbf{d} S_{d}[A] \cdot \alpha=\langle-\mathrm{d} \alpha \wedge * \mathrm{~d} A+\alpha \wedge \mathscr{J}, K\rangle=\langle\alpha \wedge(-\mathrm{d} * \mathrm{~d} A+\mathscr{J}), K\rangle .$</td><td>$\mathbf{d}S_d[A]\cdot\alpha=\langle-\mathrm{d}\alpha\wedge*\mathrm{d}A+\alpha\wedge\mathscr{J},K\rangle=\langle\alpha\wedge(-\mathrm{d}*\mathrm{d}A+\mathscr{J}),K\rangle.$</td></tr>
<tr><td>0707.4470_EQ00050</td><td>030</td><td>■ 1.000
● N|-2</td><td><code>\mathbf{d} S_{d}[A] \cdot \alpha=\langle\alpha \wedge(-\mathrm{d} * \mathrm{~d} A+\mathscr{J}), K\rangle+\langle\alpha \wedge * d A, \partial K\rangle .</code></td><td>$\mathbf{d} S_{d}[A] \cdot \alpha=\langle\alpha \wedge(-\mathrm{d} * \mathrm{~d} A+\mathscr{J}), K\rangle+\langle\alpha \wedge * d A, \partial K\rangle .$</td><td>$\mathbf{d}S_d[A]\cdot\alpha=\langle\alpha\wedge(-\mathrm{d}*\mathrm{d}A+\mathscr{J}),K\rangle+\langle\alpha\wedge*dA,\partial K\rangle.$</td></tr>
<tr><td>0707.4470_EQ00003</td><td>006</td><td>■ 1.000
● N|-1</td><td><code>\oint_{C} \mathbf{H} \cdot \mathrm{~d} \mathbf{l}=\frac{ \mathrm{d}}{\mathrm{~d} t} \int_{S} \mathbf{D} \cdot \mathrm{~d} \mathbf{A},</code></td><td>$\oint_{C} \mathbf{H} \cdot \mathrm{~d} \mathbf{l}=\frac{ \mathrm{d}}{\mathrm{~d} t} \int_{S} \mathbf{D} \cdot \mathrm{~d} \mathbf{A},$</td><td>$\oint_C \mathbf{H}\cdot\mathrm{d}\mathbf{l}=\frac{\mathrm{d}}{\mathrm{d}t}\int_S \mathbf{D}\cdot\mathrm{d}\mathbf{A},$</td></tr>
<tr><td>0707.4470_EQ00004</td><td>006</td><td>■ 1.000
● K|+0</td><td><code>\mathbf{D}=\epsilon \mathbf{E}, \quad \mathbf{B}=\mu \mathbf{H} .</code></td><td>$\mathbf{D}=\epsilon \mathbf{E}, \quad \mathbf{B}=\mu \mathbf{H} .$</td><td>$\mathbf{D}=\epsilon\mathbf{E},\qquad \mathbf{B}=\mu\mathbf{H}.$</td></tr>
<tr><td>0707.4470_EQ00005</td><td>006</td><td>■ 1.000
● K|+0</td><td><code>F=E \wedge \mathrm{~d} t+B .</code></td><td>$F=E \wedge \mathrm{~d} t+B .$</td><td>$F=E\wedge\mathrm{d}t+B.$</td></tr>
<tr><td>0707.4470_EQ00006</td><td>007</td><td>■ 1.000
● K|+0</td><td><code>G=* F=H \wedge \mathrm{~d} t-D,</code></td><td>$G=* F=H \wedge \mathrm{~d} t-D,$</td><td>$G=*F=H\wedge\mathrm{d}t-D,$</td></tr>
<tr><td>0707.4470_EQ00007</td><td>007</td><td>■ 1.000
● K|+0</td><td><code>\mathscr{J}=J \wedge \mathrm{~d} t-\rho .</code></td><td>$\mathscr{J}=J \wedge \mathrm{~d} t-\rho .$</td><td>$\mathscr{J}=J\wedge\mathrm{d}t-\rho.$</td></tr>
<tr><td>0707.4470_
EQ0011 (2.1)</td><td>007</td><td>■ 1.000
● N|-1</td><td><code>\mathrm{d} * \mathrm{~d} A=\mathscr{J} .</code></td><td>$\mathrm{d} * \mathrm{~d} A=\mathscr{J} .$</td><td>$\mathrm{d}*\mathrm{d}A=\mathscr{J}.$</td></tr>
<tr><td>0707.4470_
EQ0014 (2.6)</td><td>008</td><td>■ 1.000
● K|+0</td><td><code>\begin{aligned} \nabla \times \mathbf{H}-\partial_{t} \mathbf{D} & =\mathbf{J} \\ \nabla \cdot \mathbf{D} & = \rho . \end{aligned}</code></td><td>$\begin{aligned} \nabla \times \mathbf{H}-\partial_{t} \mathbf{D} & =\mathbf{J} \\ \nabla \cdot \mathbf{D} & = \rho . \end{aligned}$</td><td>$\begin{aligned}\nabla\times\mathbf{H}-\partial_t\mathbf{D}&=\mathbf{J}\\ \nabla\cdot\mathbf{D}&=\rho.\end{aligned}$</td></tr>
<tr><td>0707.4470_
EQ0017 (2.2)</td><td>008</td><td>■ 1.000
● N|-1</td><td><code>\mathrm{d} G=\mathscr{J}, \quad G=* F, \quad F=\mathrm{d} A .</code></td><td>$\mathrm{d} G=\mathscr{J}, \quad G=* F, \quad F=\mathrm{d} A .$</td><td>$\mathrm{d}G=\mathscr{J},\qquad G=*F,\qquad F=\mathrm{d}A.$</td></tr>
<tr><td>0707.4470_EQ00018</td><td>009</td><td>■ 1.000
● K|+0</td><td><code>\partial_{t}(\nabla \cdot \mathbf{B})=0, \quad \partial_{t}(\nabla \cdot \mathbf{D})+\nabla \cdot \mathbf{J}=\partial_{t}(\nabla \cdot \mathbf{D}-\rho)=0 .</code></td><td>$\partial_{t}(\nabla \cdot \mathbf{B})=0, \quad \partial_{t}(\nabla \cdot \mathbf{D})+\nabla \cdot \mathbf{J}=\partial_{t}(\nabla \cdot \mathbf{D}-\rho)=0 .$</td><td>$\partial_t(\nabla\cdot\mathbf{B})=0,\qquad \partial_t(\nabla\cdot\mathbf{D})+\nabla\cdot\mathbf{J}=\partial_t(\nabla\cdot\mathbf{D}-\rho)=0.$</td></tr>
<tr><td>0707.4470_EQ00019</td><td>012</td><td>■ 1.000
● N|+0</td><td><code>\langle\alpha, \sigma\rangle \equiv \int_{\sigma} \alpha .</code></td><td>$\langle\alpha, \sigma\rangle \equiv \int_{\sigma} \alpha .$</td><td>$\langle\alpha,\sigma\rangle\equiv\int_\sigma\alpha.$</td></tr>
<tr><td>0707.4470_EQ00021</td><td>012</td><td>■ 1.000
● K|+0</td><td><code>\langle\mathrm{d} \alpha, \sigma\rangle=\langle\alpha, \partial \sigma\rangle,</code></td><td>$\langle\mathrm{d} \alpha, \sigma\rangle=\langle\alpha, \partial \sigma\rangle,$</td><td>$\langle\mathrm{d}\alpha,\sigma\rangle=\langle\alpha,\partial\sigma\rangle,$</td></tr>
<tr><td>0707.4470_EQ00022</td><td>013</td><td>■ 1.000
● K|+0</td><td><code>\frac{1}{|* \sigma|}\langle * \alpha, * \sigma\rangle= \kappa(\sigma) \frac{1}{|\sigma|}\langle\alpha, \sigma \rangle,</code></td><td>$\frac{1}{|* \sigma|}\langle * \alpha, * \sigma\rangle= \kappa(\sigma) \frac{1}{|\sigma|}\langle\alpha, \sigma \rangle,$</td><td>$\frac{1}{|*\sigma|}\langle*\alpha,*\sigma\rangle=\kappa(\sigma)\frac{1}{|\sigma|}\langle\alpha,\sigma\rangle,$</td></tr>
<tr><td>0707.4470_
EQ0024 (3.1)</td><td>016</td><td>■ 1.000
● N|+0</td><td><code>(\mathrm{d} \alpha, \beta)=(\alpha, \delta \beta)+\langle \alpha \wedge * \beta, \partial K\rangle .</code></td><td>$(\mathrm{d} \alpha, \beta)=(\alpha, \delta \beta)+\langle \alpha \wedge * \beta, \partial K\rangle .$</td><td>$(\mathrm{d}\alpha,\beta)=(\alpha,\delta\beta)+\langle\alpha\wedge*\beta,\partial K\rangle.$</td></tr>
<tr><td>0707.4470_EQ00027</td><td>018</td><td>■ 1.000
● N|+0</td><td><code>\mathrm{d} F=0, \quad \mathrm{~d} G=\mathscr{J},</code></td><td>$\mathrm{d} F=0, \quad \mathrm{~d} G=\mathscr{J},$</td><td>$\mathrm{d}F=0,\qquad \mathrm{d}G=\mathscr{J},$</td></tr>
<tr><td>0707.4470_EQ00034</td><td>019</td><td>■ 1.000
● K|+0</td><td><code>\partial_{t} \mathbf{B}=-\nabla \times \mathbf{E}, \quad \nabla \cdot \mathbf{B}=0 .</code></td><td>$\partial_{t} \mathbf{B}=-\nabla \times \mathbf{E}, \quad \nabla \cdot \mathbf{B}=0 .$</td><td>$\partial_t\mathbf{B}=-\nabla\times\mathbf{E},\qquad \nabla\cdot\mathbf{B}=0.$</td></tr>
<tr><td>0707.4470_EQ00037</td><td>020</td><td>■ 1.000
● K|+0</td><td><code>\partial_{t} \mathbf{D}=\nabla \times \mathbf{H}- \mathbf{J}, \quad \nabla \cdot \mathbf{D}=\rho .</code></td><td>$\partial_{t} \mathbf{D}=\nabla \times \mathbf{H}- \mathbf{J}, \quad \nabla \cdot \mathbf{D}=\rho .$</td><td>$\partial_t\mathbf{D}=\nabla\times\mathbf{H}-\mathbf{J},\qquad \nabla\cdot\mathbf{D}=\rho.$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0707.4470_EQ0038	021	1.000 K|+0	`F=E \wedge \mathrm{~d} t+B .`	$F=E \wedge \mathrm{~d} t+B .$	$F=E\wedge \mathrm{d}t+B.$
0707.4470_EQ0039	022	1.000 N|+0	`\frac{B^{n+1}-B^{n}}{\Delta t}=-\mathrm{d}_{1} E^{n+1 / 2} .`	$\frac{B^{n+1}-B^{n}}{\Delta t}=-\mathrm{d}_{1} E^{n+1 / 2} .$	$\frac{B^{n+1}-B^{n}}{\Delta t}=-\mathrm{d}_{1} E^{n+1/2}.$
0707.4470_EQ0040	022	1.000 W|+0	`\frac{D^{n+1 / 2}-D^{n-1 / 2}}{\Delta t}= \mathrm{d}_{1}^{T} H^{n}-J^{n} .`	$\frac{D^{n+1 / 2}-D^{n-1 / 2}}{\Delta t}= \mathrm{d}_{1}^{T} H^{n}-J^{n} .$	$\frac{D^{n+1/2}-D^{n-1/2}}{\Delta t}=\mathrm{d}_{1}^{T} H^{n}-J^{n}.$
0707.4470_EQ0041	023	1.000 K|+0	`\Theta_{\sigma}=\left\{t_{\sigma}^{0}<\cdots<t_{\sigma}^{N_{\sigma}}\right\} .`	$\Theta_{\sigma}=\left\{t_{\sigma}^{0}<\cdots<t_{\sigma}^{N_{\sigma}}\right\} .$	$\Theta_{\sigma}=\left\{t_{\sigma}^{0}<\cdots<t_{\sigma}^{N_{\sigma}}\right\}.$
0707.4470_EQ0042	023	1.000 N|+0	`\Theta_{e}=\bigcup_{\sigma \ni e} \Theta_{\sigma}=\left\{t_{e}^{0} \leq \cdots \leq t_{e}^{N_{e}}\right\} .`	$\Theta_{e}=\bigcup_{\sigma \ni e} \Theta_{\sigma}=\left\{t_{e}^{0} \leq \cdots \leq t_{e}^{N_{e}}\right\} .$	$\Theta_{e}=\bigcup_{\sigma\ni e}\Theta_{\sigma}=\left\{t_{e}^{0}\leq\cdots\leq t_{e}^{N_{e}}\right\}.$
0707.4470_EQ0043	023	1.000 N|+0	`\Theta_{e}^{\prime}=\left\{t_{e}^{1 / 2} \leq \cdots \leq t_{e}^{N_{e}-1 / 2}\right\},`	$\Theta_{e}^{\prime}=\left\{t_{e}^{1 / 2} \leq \cdots \leq t_{e}^{N_{e}-1 / 2}\right\},$	$\Theta_{e}^{\prime}=\left\{t_{e}^{1/2}\leq\cdots\leq t_{e}^{N_{e}-1/2}\right\},$
0707.4470_EQ0044	024	1.000 N|+0	`\Theta_{* \sigma}=\Theta_{\sigma}, \quad \Theta_{* e}= \Theta_{e} .`	$\Theta_{* \sigma}=\Theta_{\sigma}, \quad \Theta_{* e}= \Theta_{e} .$	$\Theta_{*\sigma}=\Theta_{\sigma}, \qquad \Theta_{*e}=\Theta_{e}.$
0707.4470_EQ0045 (4.3)	024	1.000 W|+0	`\frac{B_{\sigma}^{n+1}-B_{\sigma}^{n}}{t_{\sigma}^{n+1}-t_{\sigma}^{n}}=-\mathrm{d}_{1} \sum\left\{E_{e}^{m+1 / 2}: t_{\sigma}^{n}<t_{e}^{m+1 / 2}<t_{\sigma}^{n+1}\right\} .`	$\frac{B_{\sigma}^{n+1}-B_{\sigma}^{n}}{t_{\sigma}^{n+1}-t_{\sigma}^{n}}=-\mathrm{d}_{1} \sum\left\{E_{e}^{m+1 / 2}: t_{\sigma}^{n}<t_{e}^{m+1 / 2}<t_{\sigma}^{n+1}\right\} .$	$\frac{B_{\sigma}^{n+1}-B_{\sigma}^{n}}{t_{\sigma}^{n+1}-t_{\sigma}^{n}}=-\mathrm{d}_{1}\sum\left\{E_{e}^{m+1/2}: t_{\sigma}^{n}<t_{e}^{m+1/2}<t_{\sigma}^{n+1}\right\}.$
0707.4470_EQ0048	029	1.000 N|+0	`S_{d}[A]=\left\langle\mathscr{L}_{d}, K\right\rangle .`	$S_{d}[A]=\left\langle\mathscr{L}_{d}, K\right\rangle .$	$S_{d}[A]=\left\langle\mathscr{L}_{d}, K\right\rangle.$
0707.4470_EQ0051 (5.1)	030	1.000 N|-1	`\mathbf{d} S_{d}(A) \cdot \alpha=\langle\alpha \wedge * \mathrm{~d} A, \partial K\rangle`	$\mathbf{d} S_{d}(A) \cdot \alpha=\langle\alpha \wedge * \mathrm{~d} A, \partial K\rangle$	$\mathbf{d}S_{d}(A)\cdot\alpha=\langle\alpha\wedge*\mathrm{d}A,\partial K\rangle$
0707.4470_EQ0052	030	1.000 N|-1	`\theta_{\mathscr{L}_{d}} \cdot \alpha=\alpha \wedge * \mathrm{~d} A .`	$\theta_{\mathscr{L}_{d}} \cdot \alpha=\alpha \wedge * \mathrm{~d} A .$	$\theta_{\mathscr{L}_{d}}\cdot\alpha=\alpha\wedge*\mathrm{d}A.$
0707.4470_EQ0053	030	1.000 N|+0	`\mathbf{d}^{2} S_{d}[A] \cdot \alpha \cdot \beta=\langle \mathbf{d} \theta \cdot \alpha \cdot \beta, \partial K \rangle .`	$\mathbf{d}^{2} S_{d}[A] \cdot \alpha \cdot \beta=\langle \mathbf{d} \theta \cdot \alpha \cdot \beta, \partial K \rangle .$	$\mathbf{d}^{2}S_{d}[A]\cdot\alpha\cdot\beta=\left\langle\mathbf{d}\theta\cdot\alpha\cdot\beta,\partial K\right\rangle.$
0707.4470_EQ0054 (5.2)	030	1.000 K|+0	`\left\langle\omega_{\mathscr{L}_{d}} \cdot \alpha \cdot \beta, \partial K\right\rangle=0`	$\left\langle\omega_{\mathscr{L}_{d}} \cdot \alpha \cdot \beta, \partial K\right\rangle=0$	$\left\langle\omega_{\mathscr{L}_{d}}\cdot\alpha\cdot\beta,\partial K\right\rangle=0$
0707.4470_EQ0055	031	1.000 N|+0	`A=A_{x} \mathrm{~d} x+A_{y} \mathrm{~d} y+A_{z} \mathrm{~d} z .`	$A=A_{x} \mathrm{~d} x+A_{y} \mathrm{~d} y+A_{z} \mathrm{~d} z .$	$A=A_{x}\,\mathrm{d}x+A_{y}\,\mathrm{d}y+A_{z}\,\mathrm{d}z.$
0707.4470_EQ0056	032	1.000 K|+0	`\mathrm{d}_{t} A=E \wedge \mathrm{~d} t, \quad \mathrm{~d}_{s} A=B .`	$\mathrm{d}_{t} A=E \wedge \mathrm{~d} t, \quad \mathrm{~d}_{s} A=B .$	$\mathrm{d}_{t}A=E\wedge\mathrm{d}t, \qquad \mathrm{d}_{s}A=B.$
0707.4470_EQ0057	032	1.000 N|+0	`\begin{aligned} \mathscr{L} & =-\frac{1}{2}\left(\mathrm{~d}_{t} A+\mathrm{d}_{s} A\right) \wedge *\left(\mathrm{~d}_{t} A+\mathrm{d}_{s} A\right)+A \wedge \mathscr{J} \\ & =-\frac{1}{2}\left(\mathrm{~d}_{t} A \wedge * \mathrm{~d}_{t} A+\mathrm{d}_{s} A \wedge * \mathrm{~d}_{s} A\right)+A \wedge J \wedge \mathrm{~d} t \end{aligned}`	$\begin{aligned} \mathscr{L} & =-\frac{1}{2}\left(\mathrm{~d}_{t} A+\mathrm{d}_{s} A\right) \wedge *\left(\mathrm{~d}_{t} A+\mathrm{d}_{s} A\right)+A \wedge \mathscr{J} \\ & =-\frac{1}{2}\left(\mathrm{~d}_{t} A \wedge * \mathrm{~d}_{t} A+\mathrm{d}_{s} A \wedge * \mathrm{~d}_{s} A\right)+A \wedge J \wedge \mathrm{~d} t \end{aligned}$	$\begin{aligned}\mathscr{L}&=-\frac{1}{2}(\mathrm{d}_{t}A+\mathrm{d}_{s}A)\wedge*(\mathrm{d}_{t}A+\mathrm{d}_{s}A)+A\wedge\mathscr{J}\\&=-\frac{1}{2}(\mathrm{d}_{t}A\wedge*\mathrm{d}_{t}A+\mathrm{d}_{s}A\wedge*\mathrm{d}_{s}A)+A\wedge J\wedge\mathrm{d}t\end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0707.4470_EQ0061	032	■ 1.000 ● K|+0	`\mathbf{d} S[A] \cdot \mathrm{d}_{s} f=\int_{X} \mathrm{~d}_{s} f \wedge J \wedge \mathrm{~d} t=-\int_{X} f \wedge \mathrm{~d}_{s} J \wedge \mathrm{~d} t=-\int_{X} f \wedge \mathrm{~d}_{t} \rho=-\left.\int_{\Sigma} f \wedge \rho\right|_{t_{0}} ^{t_{f}} .`	$\mathbf{d} S[A] \cdot \mathrm{d}_{s} f=\int_{X} \mathrm{~d}_{s} f \wedge J \wedge \mathrm{~d} t=-\int_{X} f \wedge \mathrm{~d}_{s} J \wedge \mathrm{~d} t=-\int_{X} f \wedge \mathrm{~d}_{t} \rho=-\left.\int_{\Sigma} f \wedge \rho\right\|_{t_{0}} ^{t_{f}} .$	$\mathbf{d} S[A] \cdot \mathrm{d}_{s} f=\int_{X} \mathrm{d}_{s} f \wedge J \wedge \mathrm{d} t=-\int_{X} f \wedge \mathrm{d}_{s} J \wedge \mathrm{d} t=-\int_{X} f \wedge \mathrm{d}_{t} \rho=-\left.\int_{\Sigma} f \wedge \rho\right\|_{t_{0}}^{t_{f}} .$
0707.4470_EQ0062	033	■ 1.000 ● K|+0	`\left.\left(\mathrm{d}_{s} D-\rho\right)\right|_{t_{0}} ^{t_{f}}=0 .`	$\left.\left(\mathrm{d}_{s} D-\rho\right)\right\|_{t_{0}} ^{t_{f}}=0 .$	$\left.\left(\mathrm{d}_{s} D-\rho\right)\right\|_{t_{0}}^{t_{f}}=0.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
0707.4470_F00001	001	—	`\nabla \cdot \mathbf{D}=\rho`	$\nabla \cdot \mathbf{D}=\rho$
0707.4470_F00002	002	—	`\mathbf{E}=-\nabla \phi`	$\mathbf{E}=-\nabla \phi$
0707.4470_F00003	003	—	`\mathbf{B}=\nabla \times \mathbf{A}`	$\mathbf{B}=\nabla \times \mathbf{A}$
0707.4470_F00004	003	—	`\mathbf{B}`	$\mathbf{B}$
0707.4470_F00005	003	—	`\nabla \cdot \nabla \times=0`	$\nabla \cdot \nabla \times=0$
0707.4470_F00006	003	—	`\nabla \times \nabla=0`	$\nabla \times \nabla=0$
0707.4470_F00007	003	—	`\mathbf{A} \mapsto \mathbf{A}+\nabla f`	$\mathbf{A} \mapsto \mathbf{A}+\nabla f$
0707.4470_F00008	003	—	`\nabla \cdot \mathbf{D}-\rho`	$\nabla \cdot \mathbf{D}-\rho$
0707.4470_F00009	003	—	`\phi`	ϕ
0707.4470_F00010	003	—	`\mathrm{d}^{2}=0`	$\mathrm{d}^{2}=0$
0707.4470_F00011	004	—	`F`	F
0707.4470_F00012	004	—	`G=* F`	$G=* F$
0707.4470_F00013	005	—	`E`	E
0707.4470_F00014	005	—	`B`	B
0707.4470_F00015	006	—	`\mathbf{E}=\left(E_{x}, E_{y}, E_{z}\right)`	$\mathbf{E}=\left(E_{x}, E_{y}, E_{z}\right)$
0707.4470_F00016	006	—	`\mathbf{B}=\left(B_{x}, B_{y}, B_{z}\right)`	$\mathbf{B}=\left(B_{x}, B_{y}, B_{z}\right)$
0707.4470_F00017	006	—	`\mathbf{E}`	$\mathbf{E}$
0707.4470_F00018	006	—	`H`	H
0707.4470_F00019	006	—	`D`	D
0707.4470_F00020	006	—	`\epsilon`	ϵ
0707.4470_F00021	006	—	`\mu`	μ
0707.4470_F00022	006	—	`*_{\epsilon}`	$*_{\epsilon}$
0707.4470_F00023	006	—	`*_{\mu}`	$*_{\mu}$
0707.4470_F00024	006	—	`\epsilon=\epsilon_{0}`	$\epsilon=\epsilon_{0}$
0707.4470_F00025	006	—	`\mu=\mu_{0}`	$\mu=\mu_{0}$
0707.4470_F00026	006	—	`\epsilon_{0}=\mu_{0}=c=1`	$\epsilon_{0}=\mu_{0}=c=1$
0707.4470_F00027	006	—	`\rho \mathrm{d} x \wedge \mathrm{~d} y \wedge \mathrm{~d} z`	$\rho \mathrm{d} x \wedge \mathrm{~d} y \wedge \mathrm{~d} z$
0707.4470_F00028	006	—	`J=J_{x} \mathrm{~d} y \wedge \mathrm{~d} z+J_{y} \mathrm{~d} z \wedge \mathrm{~d} x+J_{z} \mathrm{~d} x \wedge \mathrm{~d} y`	$J=J_{x} \mathrm{~d} y \wedge \mathrm{~d} z+J_{y} \mathrm{~d} z \wedge \mathrm{~d} x+J_{z} \mathrm{~d} x \wedge \mathrm{~d} y$
0707.4470_F00029	006	—	`\partial_{t} \rho+\mathrm{d} J=0`	$\partial_{t} \rho+\mathrm{d} J=0$
0707.4470_F00030	007	—	`\rho`	ρ
0707.4470_F00031	007	—	`J`	J
0707.4470_F00032	007	—	`\mathscr{J}`	$\mathscr{J}$
0707.4470_F00033	007	—	`\mathrm{d} \mathscr{J}=0`	$\mathrm{d} \mathscr{J}=0$
0707.4470_F00034	007	—	`A`	A
0707.4470_F00035	007	—	`F=\mathrm{d} A`	$F=\mathrm{d} A$
0707.4470_F00036	007	—	`X`	X
0707.4470_F00037	007	—	`\alpha`	α
0707.4470_F00038	007	—	`\partial X`	∂X
0707.4470_F00039	007	—	`G=* F=* \mathrm{~d} A`	$G=* F=* \mathrm{~d} A$
0707.4470_F00040	007	—	`\mathrm{d} G=\mathscr{J}`	$\mathrm{d} G=\mathscr{J}$
0707.4470_F00041	008	—	`\mathrm{d} F=\mathrm{d}^{2} A=0`	$\mathrm{d} F=\mathrm{d}^{2} A=0$
0707.4470_F00042	008	—	`(x, y, z, t)`	(x, y, z, t)
0707.4470_F00043	008	—	`X=\mathbb{R}^{3,1}`	$X=\mathbb{R}^{3,1}$
0707.4470_F00044	008	—	`F=E \wedge \mathrm{~d} t+B`	$F=E \wedge \mathrm{~d} t+B$
0707.4470_F00045	008	—	`G=* F=H \wedge \mathrm{~d} t-D`	$G=* F=H \wedge \mathrm{~d} t-D$
0707.4470_F00046	008	—	`G`	G
0707.4470_F00047	008	—	`\alpha, \phi, \gamma`	α, ϕ, γ
0707.4470_F00048	009	—	`\nabla \cdot \mathbf{B}`	$\nabla \cdot \mathbf{B}$
0707.4470_F00049	009	—	`\mathrm{d} A`	$\mathrm{d} A$
0707.4470_F00050	009	—	`A \mapsto A+\mathrm{d} f`	$A \mapsto A+\mathrm{d} f$
0707.4470_F00051	009	—	`\phi=A(\partial / \partial t)=0`	$\phi=A(\partial / \partial t)=0$
0707.4470_F00052	009	—	`\mathrm{d}^{2} A=0`	$\mathrm{d}^{2} A=0$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered
0707.4470_F00053	011	—	n	n
0707.4470_F00054	011	—	K	K
0707.4470_F00055	011	—	\sigma	σ
0707.4470_F00056	011	—	* K	$* K$
0707.4470_F00057	011	—	k	k
0707.4470_F00058	011	—	(n-k)	$(n-k)$
0707.4470_F00059	011	—	* \sigma	$* \sigma$
0707.4470_F00060	011	—	(* K	$(* K$
0707.4470_F00061	011	—	(n-1)	$(n-1)$
0707.4470_F00062	011	—	\alpha^{k}	α^{k}
0707.4470_F00063	011	—	\sigma^{k}	σ^{k}
0707.4470_F00064	011	—	\left\langle\alpha^{k}, \sigma^{k}\right\rangle	$\left\langle\alpha^{k}, \sigma^{k}\right\rangle$
0707.4470_F00065	012	—	\sigma^{1}	σ^{1}
0707.4470_F00066	012	—	* \sigma^{1}	$* \sigma^{1}$
0707.4470_F00067	012	—	\mathrm{CH}\left(\sigma^{1}, * \sigma^{1}\right)	$\mathrm{CH}\left(\sigma^{1}, * \sigma^{1}\right)$
0707.4470_F00068	012	—	\langle\cdot, \cdot\rangle	$\langle\cdot, \cdot\rangle$
0707.4470_F00069	012	—	n-k	$n-k$
0707.4470_F00070	012	—	k+1	$k+1$
0707.4470_F00071	012	—	\mathrm{d} \alpha	$\mathrm{d} \alpha$
0707.4470_F00072	012	—	(k+1)	$(k+1)$
0707.4470_F00073	012	—	\partial \sigma	$\partial \sigma$
0707.4470_F00074	013	—	* \alpha	$* \alpha$
0707.4470_F00075	013	—	|\sigma|	$\lvert\sigma\rvert$
0707.4470_F00076	013	—	|* \sigma|	$\lvert * \sigma\rvert$
0707.4470_F00077	013	—	\kappa	κ
0707.4470_F00078	013	—	\mathbb{L}^{2}	$\mathbb{L}^{2}$
0707.4470_F00079	013	—	(\cdot, \cdot)	$(\cdot, \cdot)$
0707.4470_F00080	013	—	\mathrm{CH}(\sigma, * \sigma)	$\mathrm{CH}(\sigma, * \sigma)$
0707.4470_F00081	013	—	\sigma \cup * \sigma	$\sigma \cup * \sigma$
0707.4470_F00082	013	—	|\mathrm{CH}(\sigma, * \sigma)|=\binom{n}{k}^{-1}|\sigma||* \sigma|	$\lvert\mathrm{CH}(\sigma, * \sigma)\rvert=\binom{n}{k}^{-1}\lvert\sigma\rvert\lvert * \sigma\rvert$
0707.4470_F00083	013	—	\alpha \wedge * \beta	$\alpha \wedge * \beta$
0707.4470_F00084	013	—	\delta	δ
0707.4470_F00085	013	—	\mathrm{d}, *, \delta	$\mathrm{d}, *, \delta$
0707.4470_F00086	013	—	\sigma_{i}^{k}	σ_{i}^{k}
0707.4470_F00087	013	—	\alpha_{i}=\left\langle\alpha, \sigma_{i}^{k}\right\rangle	$\alpha_{i}=\left\langle\alpha, \sigma_{i}^{k}\right\rangle$
0707.4470_F00088	014	—	k=0,1,2,3	$k=0,1,2,3$
0707.4470_F00089	014	—	*	$*$
0707.4470_F00090	014	—	*^{-1}	$*^{-1}$
0707.4470_F00091	014	—	\kappa\left(\sigma_{i}^{k}\right) \frac{\left|* \sigma_{i}^{k}\right|}{\left|\sigma_{i}^{k}\right|}	$\kappa\left(\sigma_{i}^{k}\right) \frac{\left\lvert * \sigma_{i}^{k}\right\rvert}{\left\lvert\sigma_{i}^{k}\right\rvert}$
0707.4470_F00092	014	—	\partial K	∂K
0707.4470_F00093	014	—	t_{0}	t_{0}
0707.4470_F00094	014	—	t_{0}, E	t_{0}, E
0707.4470_F00095	014	—	t_{1}	t_{1}
0707.4470_F00096	014	—	t_{1 / 2}	$t_{1 / 2}$
0707.4470_F00097	015	—	n-1	$n-1$
0707.4470_F00098	015	—	(k-1)	$(k-1)$
0707.4470_F00099	015	—	*(\partial K)	$*(\partial K)$
0707.4470_F00100	015	—	\partial(* K)=*(\partial K)	$\partial(* K)=*(\partial K)$
0707.4470_F00101	015	—	t_{-\epsilon}	$t_{-\epsilon}$
0707.4470_F00102	015	—	\epsilon \rightarrow 0	$\epsilon \rightarrow 0$
0707.4470_F00103	016	—	k-1	$k-1$
0707.4470_F00104	016	—	\beta	β
0707.4470_F00105	016	—	* \beta	$* \beta$
0707.4470_F00106	016	—	F, G	F, G

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered
0707.4470_F00107	016	—	`\mathbb{R}^{3,1}`	$\mathbb{R}^{3,1}$
0707.4470_F00108	016	—	`p`	p
0707.4470_F00109	016	—	`\Delta x, \Delta y, \Delta z, \Delta t`	$\Delta x, \Delta y, \Delta z, \Delta t$
0707.4470_F00110	017	—	`E_{x} \Delta x \Delta t`	$E_{x} \Delta x \Delta t$
0707.4470_F00111	017	—	`x t`	$x t$
0707.4470_F00112	017	—	`E_{y} \Delta y \Delta t`	$E_{y} \Delta y \Delta t$
0707.4470_F00113	017	—	`y t`	$y t$
0707.4470_F00114	017	—	`E_{z} \Delta z \Delta t`	$E_{z} \Delta z \Delta t$
0707.4470_F00115	017	—	`z t`	$z t$
0707.4470_F00116	017	—	`B_{x} \Delta y \Delta z`	$B_{x} \Delta y \Delta z$
0707.4470_F00117	017	—	`y z`	$y z$
0707.4470_F00118	017	—	`B_{y} \Delta z \Delta x`	$B_{y} \Delta z \Delta x$
0707.4470_F00119	017	—	`x z`	$x z$
0707.4470_F00120	017	—	`B_{z} \Delta x \Delta y`	$B_{z} \Delta x \Delta y$
0707.4470_F00121	017	—	`x y`	$x y$
0707.4470_F00122	017	—	`\left[x_{k}, x_{k+1}\right] \times`	$\left[x_{k}, x_{k+1}\right] \times$
0707.4470_F00123	017	—	`\left[y_{l}, y_{l+1}\right] \times\left[z_{m}, z_{m+1}\right] \times\left[t_{n}, t_{n+1}\right]`	$\left[y_{l}, y_{l+1}\right] \times\left[z_{m}, z_{m+1}\right] \times\left[t_{n}, t_{n+1}\right]$
0707.4470_F00124	017	—	`\left.F\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}`	$\left.F\right\|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}$
0707.4470_F00125	017	—	`\left[x_{k}, x_{k+1}\right] \times\left\{y_{l}\right\} \times\left\{z_{m}\right\} \times \left[t_{n}, t_{n+1}\right]`	$\left[x_{k}, x_{k+1}\right] \times\left\{y_{l}\right\} \times\left\{z_{m}\right\} \times \left[t_{n}, t_{n+1}\right]$
0707.4470_F00126	018	—	`\mathrm{d} F`	$\mathrm{d} F$
0707.4470_F00127	020	—	`\mathbf{D}=\epsilon \mathbf{E}, \mathbf{B}=\mu \mathbf{H}`	$\mathbf{D}=\epsilon \mathbf{E}, \mathbf{B}=\mu \mathbf{H}$
0707.4470_F00128	021	—	`\kappa=1`	$\kappa=1$
0707.4470_F00129	021	—	`\kappa=-1`	$\kappa=-1$
0707.4470_F00130	021	—	`E \Delta t`	$E \Delta t$
0707.4470_F00131	021	—	`H \Delta t`	$H \Delta t$
0707.4470_F00132	021	—	`B^{n}`	B^{n}
0707.4470_F00133	021	—	`H^{n}`	H^{n}
0707.4470_F00134	021	—	`E^{n+1 / 2}`	$E^{n+1 / 2}$
0707.4470_F00135	021	—	`D^{n+1 / 2}`	$D^{n+1 / 2}$
0707.4470_F00136	021	—	`\mathrm{d}_{1}`	d_{1}
0707.4470_F00137	021	—	`\mathrm{d}_{1}^{T}`	d_{1}^{T}
0707.4470_F00138	021	—	`\mathrm{d} F=0`	$\mathrm{d} F=0$
0707.4470_F00139	023	—	`\Delta t_{\sigma}=t_{\sigma}^{n+1}-t_{\sigma}^{n}`	$\Delta t_{\sigma}=t_{\sigma}^{n+1}-t_{\sigma}^{n}$
0707.4470_F00140	023	—	`\Delta t`	Δt
0707.4470_F00141	023	—	`\Theta_{\sigma} \cap \Theta_{\sigma^{\prime}}=\left\{t_{0}\right\}`	$\Theta_{\sigma} \cap \Theta_{\sigma^{\prime}}=\left\{t_{0}\right\}$
0707.4470_F00142	023	—	`\sigma \neq \sigma^{\prime}`	$\sigma \neq \sigma^{\prime}$
0707.4470_F00143	023	—	`e`	e
0707.4470_F00144	023	—	`t_{e}^{k+1 / 2}=\left(t_{e}^{k+1}+t_{e}^{k}\right) / 2`	$t_{e}^{k+1 / 2}=\left(t_{e}^{k+1}+t_{e}^{k}\right) / 2$
0707.4470_F00145	023	—	`\Theta_{e} \supset \Theta_{\sigma}`	$\Theta_{e} \supset \Theta_{\sigma}$
0707.4470_F00146	023	—	`e \subset \sigma`	$e \subset \sigma$
0707.4470_F00147	024	—	`\nVdash_{\left\{t_{\sigma}^{n}=t_{e}^{m}\right\}}`	$\nVdash_{\left\{t_{\sigma}^{n}=t_{e}^{m}\right\}}$
0707.4470_F00148	024	—	`t_{\sigma}^{n}=t_{e}^{m}`	$t_{\sigma}^{n}=t_{e}^{m}$
0707.4470_F00149		—	`t_{\sigma}^{n+1}`	t_{σ}^{n+1}
0707.4470_F00150		—	`B_{\sigma}^{n+1}`	B_{σ}^{n+1}
0707.4470_F00151		—	`H_{\sigma}^{n+1}=*_{\mu}^{-1} B_{\sigma}^{n+1}`	$H_{\sigma}^{n+1}=*_{\mu}^{-1} B_{\sigma}^{n+1}$
0707.4470_F00152		—	`D_{e}^{m+3 / 2}`	$D_{e}^{m+3 / 2}$
0707.4470_F00153		—	`E_{e}^{m+3 / 2}=*_{\epsilon}^{-1} D_{e}^{m+3 / 2}`	$E_{e}^{m+3 / 2}=*_{\epsilon}^{-1} D_{e}^{m+3 / 2}$
0707.4470_F00154	024	—	`\mathbf{x}`	$\mathbf{x}$
0707.4470_F00155	024	—	`\dot{\mathbf{x}}`	$\dot{\mathbf{x}}$
0707.4470_F00156		—	`A_{e} \leftarrow A_{e}^{0}, E_{e} \leftarrow E_{e}^{1 / 2}, \tau_{e} \leftarrow t_{0} / /`	$A_{e} \leftarrow A_{e}^{0}, E_{e} \leftarrow E_{e}^{1 / 2}, \tau_{e} \leftarrow t_{0} / /$
0707.4470_F00157		—	`\tau_{\sigma} \leftarrow t_{0}`	$\tau_{\sigma} \leftarrow t_{0}$
0707.4470_F00158		—	`t_{\sigma}^{1}`	t_{σ}^{1}
0707.4470_F00159		—	`Q . \mathrm{push}\left(t_{\sigma}^{1}, \sigma\right) / /`	$Q . \mathrm{push}\left(t_{\sigma}^{1}, \sigma\right) / /$
0707.4470_F00160		—	`(t, \sigma) \leftarrow Q . \operatorname{pop}() / /`	$(t, \sigma) \leftarrow Q . \operatorname{pop}() / /$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

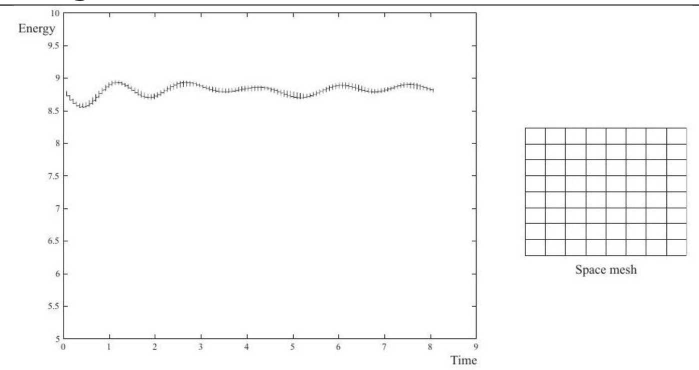
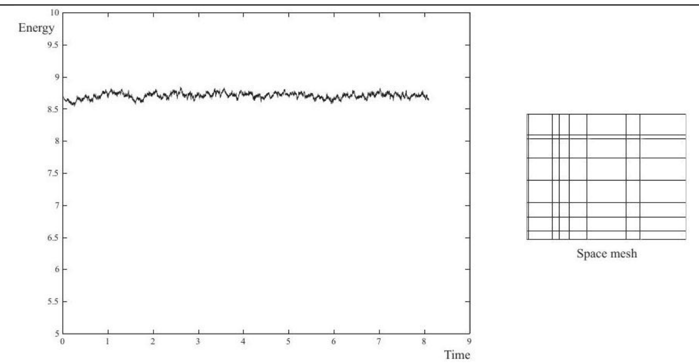
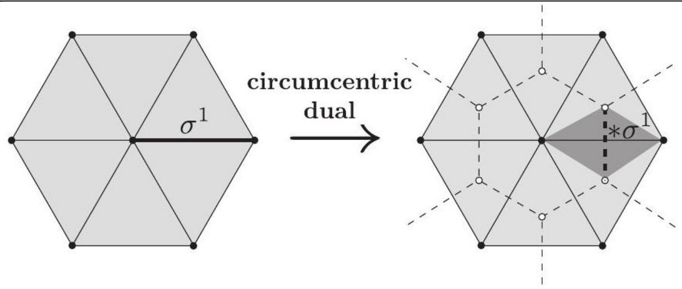
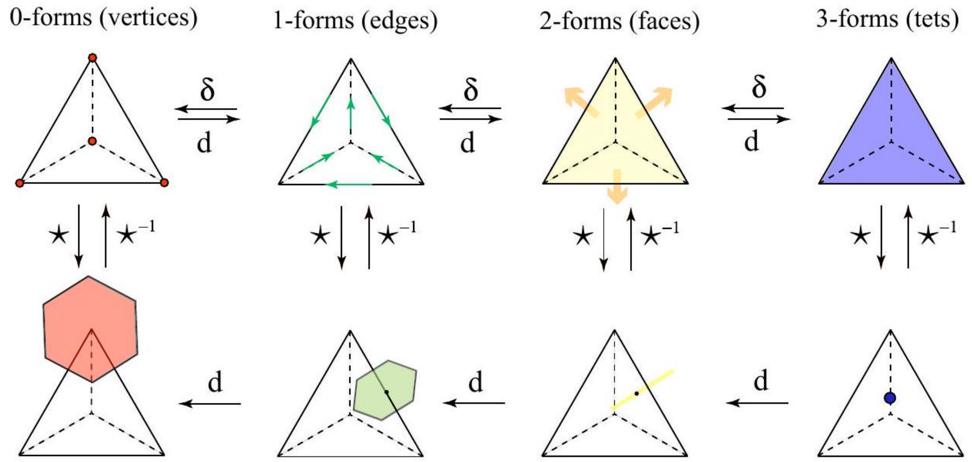
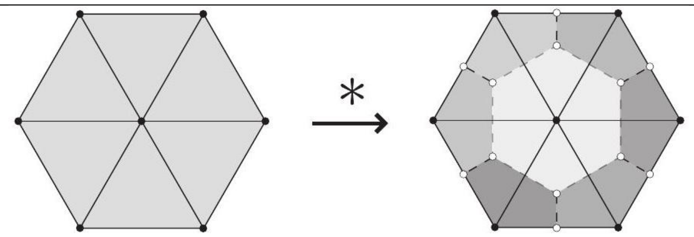
Identifier	Page	Conf.	LaTeX source	Rendered
0707.4470_F00161		—	t	t
0707.4470_F00162		—	A_{e} \leftarrow A_{e}-E_{e}\left(t-\tau_{e}\right) / /	$A_{e} \leftarrow A_{e}-E_{e}\left(t-\tau_{e}\right) / /$
0707.4470_F00163		—	t<	$t<$
0707.4470_F00164		—	B_{\sigma} \leftarrow \mathrm{d}_{1} A_{e}	$B_{\sigma} \leftarrow \mathrm{d}_{1} A_{e}$
0707.4470_F00165		—	H_{\sigma} \leftarrow *_{\mu} B_{\sigma}	$H_{\sigma} \leftarrow *_{\mu} B_{\sigma}$
0707.4470_F00166		—	D_{e} \leftarrow *_{\epsilon} E_{e}	$D_{e} \leftarrow *_{\epsilon} E_{e}$
0707.4470_F00167		—	D_{e} \leftarrow D_{e}+\mathrm{d}_{1}(e, \sigma) H_{\sigma}\left(t-\tau_{\sigma}\right)	$D_{e} \leftarrow D_{e}+\mathrm{d}_{1}(e, \sigma) H_{\sigma}\left(t-\tau_{\sigma}\right)$
0707.4470_F00168		—	E_{e} \leftarrow *_{\epsilon} D_{e}	$E_{e} \leftarrow *_{\epsilon} D_{e}$
0707.4470_F00169		—	\tau_{\sigma} \leftarrow t / /	$\tau_{\sigma} \leftarrow t / /$
0707.4470_F00170		—	t_{\sigma}^{\text {next }}	$t_{\sigma}^{\text {next }}$
0707.4470_F00171		—	Q . \mathrm{push}\left(t_{\sigma}^{\text {next }}, \sigma\right) / /	$Q . \mathrm{push}\left(t_{\sigma}^{\text {next }}, \sigma\right) / /$
0707.4470_F00172	025	—	x	x
0707.4470_F00173	025	—	y	y
0707.4470_F00174	027	—	3+1	$3+1$
0707.4470_F00175 **[refined: source]**	029		\mathcal{J}	$\mathcal{J}$
0707.4470_F00176	029	—	\mathrm{d} * \mathrm{~d} A=\mathscr{J}	$\mathrm{d} * \mathrm{~d} A=\mathscr{J}$
0707.4470_F00177		—	\square	$\square$
0707.4470_F00178	029	—	\theta_{\mathscr{L}}	$\theta_{\mathscr{L}}$
0707.4470_F00179	029	—	(n+1)	$(n+1)$
0707.4470_F00180	029	—	(n+2)	$(n+2)$
0707.4470_F00181	029	—	\omega_{\mathscr{L}}	$\omega_{\mathscr{L}}$
0707.4470_F00182	029	—	\omega_{\mathscr{L}}=-\mathbf{d} \theta_{\mathscr{L}}	$\omega_{\mathscr{L}}=-\mathbf{d} \theta_{\mathscr{L}}$
0707.4470_F00183	029	—	\mathbf{d}^{2}=0	$\mathbf{d}^{2}=0$
0707.4470_F00184	029	—	n=0	$n=0$
0707.4470_F00185	029	—	\omega_{L}	ω_{L}
0707.4470_F00186	030	—	K \subset \mathscr{K}	$K \subset \mathscr{K}$
0707.4470_F00187	030	—	S_{d}	S_{d}
0707.4470_F00188	030	—	\theta_{\mathscr{L}_{d}}	$\theta_{\mathscr{L}_{d}}$
0707.4470_F00189	030	—	\omega_{\mathscr{L}_{d}}=-\mathbf{d} \theta_{\mathscr{L}_{d}}	$\omega_{\mathscr{L}_{d}}=-\mathbf{d} \theta_{\mathscr{L}_{d}}$
0707.4470_F00190	030	—	\mathbf{d}^{2} S_{d}=0	$\mathbf{d}^{2} S_{d}=0$
0707.4470_F00191	030	—	\alpha, \beta	α, β
0707.4470_F00192	031	—	\mathscr{K}	$\mathscr{K}$
0707.4470_F00193	031	—	\omega_{\mathscr{L}_{d}}	$\omega_{\mathscr{L}_{d}}$
0707.4470_F00194	031	—	\omega_{\mathscr{L}_{d}} \cdot \alpha \cdot \beta	$\omega_{\mathscr{L}_{d}} \cdot \alpha \cdot \beta$
0707.4470_F00195	031	—	\rho=\nabla \cdot \mathbf{D}	$\rho=\nabla \cdot \mathbf{D}$
0707.4470_F00196	031	—	f	f
0707.4470_F00197	031	—	A_{t}=0	$A_{t}=0$
0707.4470_F00198	031	—	A_{x}, A_{y}, A_{z}	A_{x}, A_{y}, A_{z}
0707.4470_F00199	031	—	\mathbf{A}	$\mathbf{A}$
0707.4470_F00200	031	—	A=\mathbf{A}^{b}	$A=\mathbf{A}^{b}$
0707.4470_F00201	031	—	\mathrm{d}_{t}	d_{t}
0707.4470_F00202	032	—	\mathrm{d}_{s}	d_{s}
0707.4470_F00203	032	—	\mathrm{d}=\mathrm{d}_{t}+\mathrm{d}_{s}	$\mathrm{d}=\mathrm{d}_{t}+\mathrm{d}_{s}$
0707.4470_F00204	032	—	\mathrm{d} t	$\mathrm{d} t$
0707.4470_F00205	032	—	\mathrm{d}_{s} A	$\mathrm{d}_{s} A$
0707.4470_F00206	032	—	\mathrm{d}_{t} A	$\mathrm{d}_{t} A$
0707.4470_F00207	032	—	\mathrm{d}_{s} D=\rho	$\mathrm{d}_{s} D=\rho$
0707.4470_F00208	032	—	t_{f}	t_{f}
0707.4470_F00209	032	—	\Sigma	Σ
0707.4470_F00210	032	—	\alpha=\mathrm{d}_{s} f	$\alpha=\mathrm{d}_{s} f$
0707.4470_F00211	033	—	\mathrm{d}_{s} D-\rho	$\mathrm{d}_{s} D-\rho$
0707.4470_F00212	033	—	\nabla \cdot \mathbf{D}	$\nabla \cdot \mathbf{D}$
0707.4470_F00213	036	—	\mathbb{R}^{3}	$\mathbb{R}^{3}$

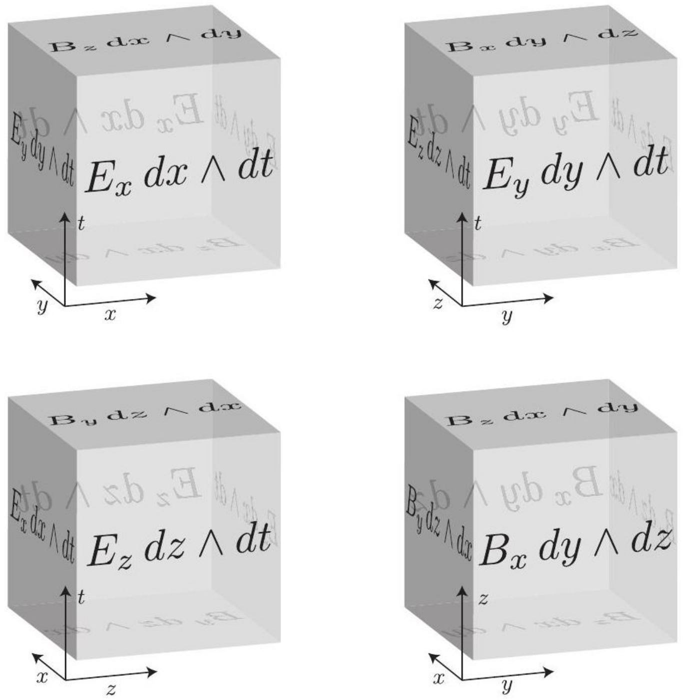
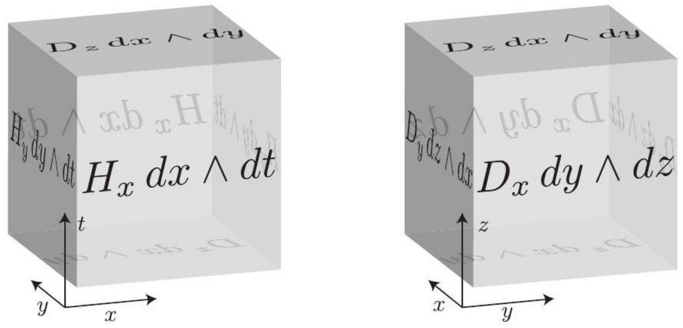
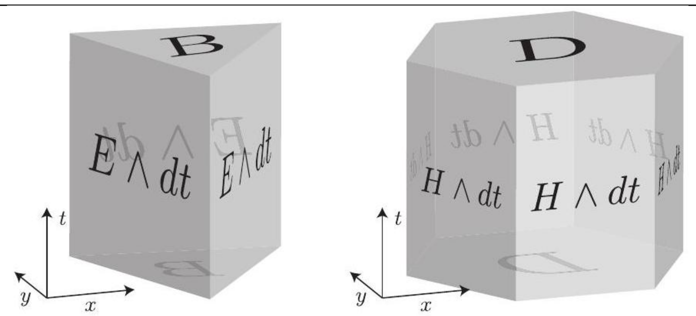
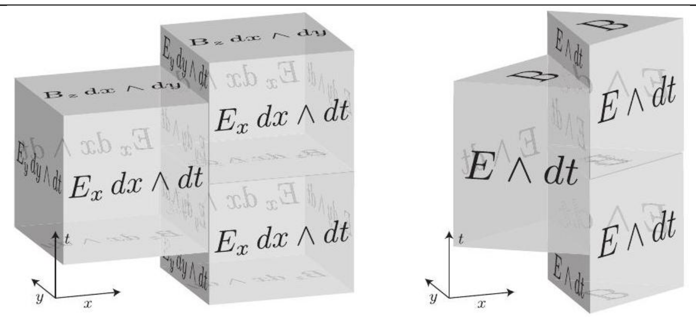
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

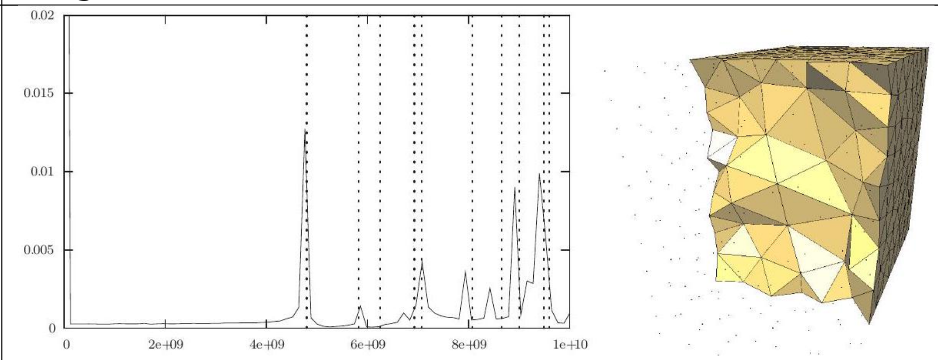
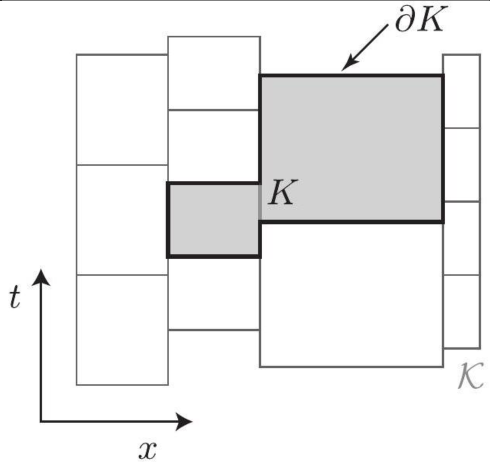
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Unrecovered image regions — TikZ / table / failed-math candidates

Regions MathPix left as images (no LaTeX). Reconstruct with `pdfdrill vision` (LLM classify + transcribe to TikZ/tabular/math); verify an LLM result against the real ink with inkdrill.

Identifier	Page	Image	Source
0707.4470_PIC_0001	026		tex.zip (local)
0707.4470_PIC_0002	026		tex.zip (local)
0707.4470_DIA_0001	012		tex.zip (local)
0707.4470_DIA_0002	014		tex.zip (local)
0707.4470_DIA_0003	015		tex.zip (local)

Identifier	Page	Image	Source
0707.4470_DIA_0004	017		tex.zip (local)
0707.4470_DIA_0005	021		tex.zip (local)
0707.4470_DIA_0006	022		tex.zip (local)
0707.4470_DIA_0007	023		tex.zip (local)
0707.4470_DIA_0008	025	*(image not on disk — pass –texzip or download the CDN crop)*	CDN only

Identifier	Page	Image	Source
0707.4470_DIA_0009	027		tex.zip (local)
0707.4470_DIA_0010	031		tex.zip (local)